

Fourier Series and Half-Wave Symmetry

1 Background Knowledge Check

1. Odd and Even Functions

- Any given function can be classified as having odd, even or neither symmetry
- Even functions are symmetrical about the y-axis and satisfy $f(-x) = f(x)$
- Odd functions are symmetrical about the origin and satisfy $f(-x) = -f(x)$
- For more information and/or to review odd and even functions, please refer to the Student Learning Services's "Odd and Even Functions" supplemental material

2. Periodic Functions

- A function is considered periodic if it repeats itself after a certain interval
- Two common examples are the sine and cosine functions, which each have a period of 2π , which means both functions start repeating themselves after an interval of 2π
- Periodic functions satisfy $f(x) = f(x+T)$ where T is the period of the function
- Since $\sin(x)$ is a periodic function with a period of 2π , it satisfies $f(x) = \sin(x) = f(x + 2\pi) = \sin(x + 2\pi)$. Check at $x = \pi$:

$$\sin(\pi) = 0$$

$$\sin(\pi + 2\pi) = \sin(3\pi) = 0$$

$$\sin(\pi) = \sin(3\pi)$$

3. Other fundamentals for Fourier Series

To begin working with fourier series, fundamental knowledge about the following topics is also required:

- Integrals
- Series
- Trigonometric Functions (sine, cosine etc.)

If you find you lack knowledge in any of the topics mentioned above, please review your course material before reading through the rest of this document.

2 Half-Wave Symmetry

Like odd and even symmetry, half-wave symmetry is related to the graphical geometry of a function. A function exhibits half-wave symmetry if one half of its wavelength is the negative of the other half of the wavelength after a shift of a half wavelength. Mathematically, a function that exhibits half-wave symmetry must satisfy:

$$f(x) = -f(x + \frac{T}{2})$$

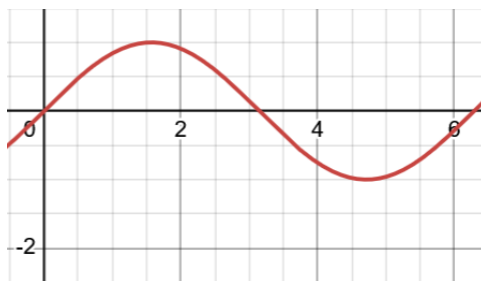
where T is the length of one period of $f(x)$.

Let's try to build our intuition by working with an example:

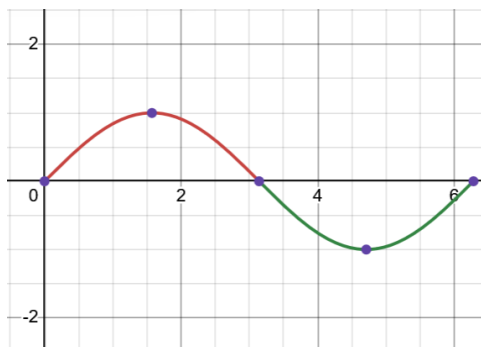
- Does $f(x) = \sin(x)$ exhibit half-wave symmetry?

1. Graphically:

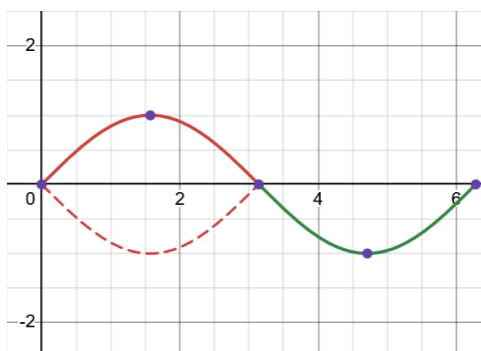
Let's consider the graph of $f(x) = \sin(x)$:



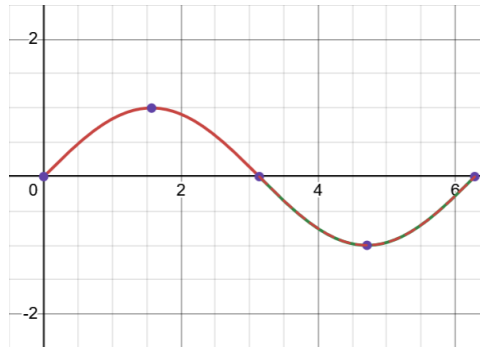
One period of $\sin(x)$ lies on the interval $[0, 2\pi]$, so the first and second halves are $[0, \pi]$ and $[\pi, 2\pi]$ respectively:



Taking the first half of the wavelength and reflecting it along the x-axis gives us the following graph:



Now, if we move this reflection right by half of a wavelength (π), it perfectly matches the second-half of our wavelength. Therefore, $f(x) = \sin(x)$ exhibits half-wave symmetry:



2. Mathematically:

$$f(x) = \sin(x), \quad T = 2\pi \Rightarrow \frac{T}{2} = \pi$$

$$-f\left(x + \frac{T}{2}\right) = -\sin(x + \pi)$$

Using the identity $\sin(x + \pi) = -\sin(x)$:

$$-\sin(x + \pi) = -(-\sin(x)) = \sin(x) = f(x)$$

Therefore, $\sin(x)$ exhibits half-wave symmetry.

3 Fourier Series

A Fourier series is an infinite series composed of terms consisting of sine and cosine functions. Its purpose is to represent or approximate a periodic function as a sum of these trigonometric functions. This is similar to how a Taylor or Maclaurin series achieves this through the use of polynomials (Don't worry if you are not familiar with Taylor or Maclaurin series!). The general formula for a Fourier series is as follows:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

or

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

where:

$f(x)$ is the function we are trying to approximate using the Fourier Series

n is the index of our summation notation

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

(When integrating for a_0 , a_n or b_n ; treat n as a constant)

Therefore, to calculate the Fourier Series for any function, we need to solve the three integrals above corresponding to a_0 , a_n and b_n and use them to form the equation for our Fourier Series.

Keeping in mind that $\cos(x)$ is an even function and $\sin(x)$ is an odd function, we can make some simplifications to our Fourier Series if we know whether or not our original function is an odd function, an even function or neither:

Fourier Series Symmetry Rules

If $f(x)$ is even: $b_n = 0 \Rightarrow$ no sine terms

sine is an odd function; therefore, we don't need it to approximate even functions

If $f(x)$ is odd: $a_n = 0$, $a_0 = 0 \Rightarrow$ no cosine terms

cosine is an even function, therefore, we don't need it to approximate odd functions

a_0 represents the average value of $f(x)$ over $[-\pi, \pi]$, which for an odd function is always 0

- **Example:** Find the Fourier Series for $f(x) = x$ on $[-\pi, \pi]$

Since x is an odd function:

$$a_0 = 0, \quad a_n = 0$$

We only need to compute b_n :

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$$

Using integration by parts, let

$$u = x, \quad dv = \sin(nx) dx$$

$$du = dx, \quad v = -\frac{\cos(nx)}{n}$$

Then

$$b_n = \frac{1}{\pi} \left[-\frac{x \cos(nx)}{n} \right]_{-\pi}^{\pi} + \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{\cos(nx)}{n} dx$$

Evaluate the boundary term:

$$\left[-\frac{x \cos(nx)}{n} \right]_{-\pi}^{\pi} = -\frac{\pi \cos(n\pi)}{n} - \frac{\pi \cos(-n\pi)}{n}$$

Since cosine is even, $\cos(-n\pi) = \cos(n\pi)$:

$$= -\frac{2\pi \cos(n\pi)}{n}$$

Now evaluate the remaining integral:

$$\begin{aligned} \frac{1}{n\pi} \int_{-\pi}^{\pi} \cos(nx) dx &= \frac{1}{n\pi} \left[\frac{\sin(nx)}{n} \right]_{-\pi}^{\pi} \\ &= \frac{1}{n\pi} \left(\frac{\sin(n\pi) - \sin(-n\pi)}{n} \right) = 0 \end{aligned}$$

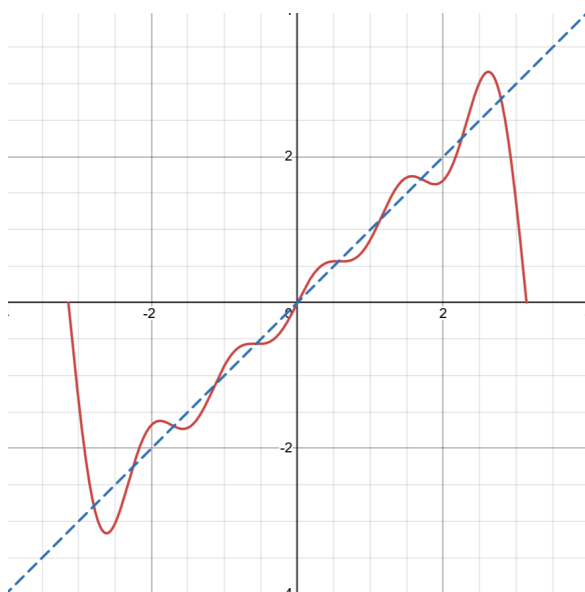
Therefore,

$$b_n = \frac{1}{\pi} \left(-\frac{2\pi \cos(n\pi)}{n} \right) = -\frac{2 \cos(n\pi)}{n}$$

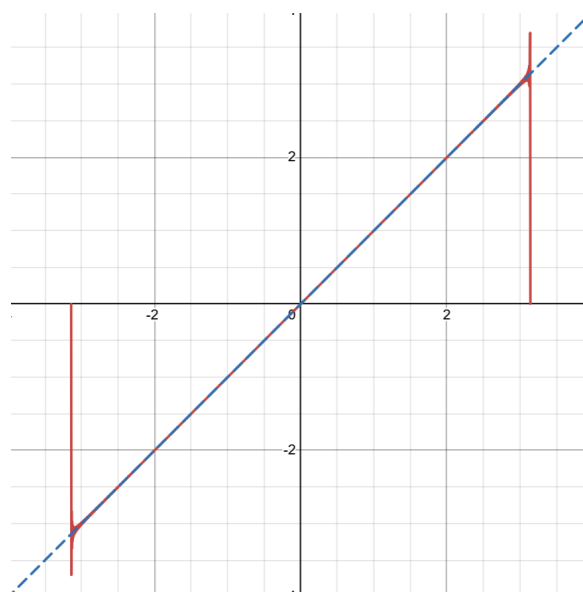
Hence, the Fourier series is

$$f(x) = x = \sum_{n=1}^{\infty} \frac{-2 \cos(n\pi)}{n} \sin(nx)$$

(as $n \rightarrow \infty$, the series converges to $f(x)$)



$n = 5$



$n = 999$

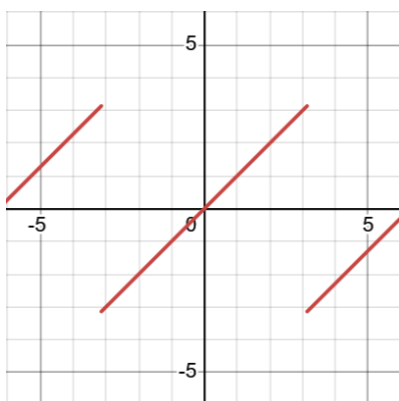
$f(x) = x$ and its Fourier Series approximations

Practice Questions

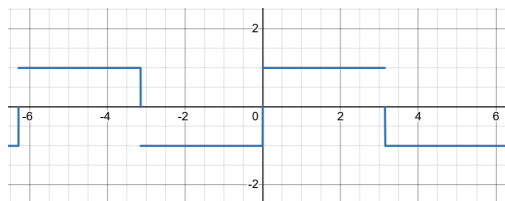
1. Find the Fourier Series for the following functions:

- (a) $f(x) = x^2$
- (b) $f(x) = x^3$
- (c) $f(x) = -3x + 4$
- (d) $f(x) = e^x$
- (e) $f(x) = \tan(x)$

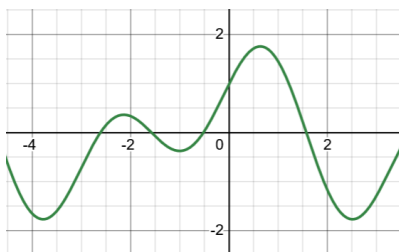
2. Determine if the following functions exhibit half-wave symmetry using their graphs:



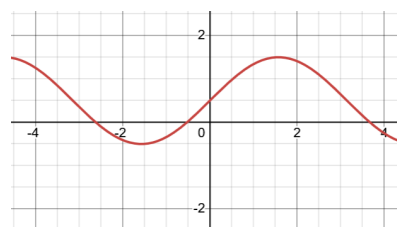
(a)



(b)



(c)



(d)

3. Determine if the following functions exhibit half-wave symmetry (you may graph the functions or check mathematically):

- (a) $f(x) = \sin(x) - 1$
 - (b) $f(x) = \cos(x + \pi)$
 - (c) $f(x) = 3 \tan(x)$
 - (d) $f(x) = \sin(\pi x) + \pi$
 - (e) $f(x) = 92 \cos(24x - 49)$
-

Solutions

1. Fourier Series on $[-\pi, \pi]$:

(a) $f(x) = x^2$ (even $\Rightarrow b_n = 0$)

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

(b) $f(x) = x^3$ (odd $\Rightarrow a_0 = 0, a_n = 0$)

$$x^3 = \sum_{n=1}^{\infty} (-1)^n \left(-\frac{2\pi^2}{n} + \frac{12}{n^3} \right) \sin(nx)$$

(c) $f(x) = -3x + 4$

Split into odd + even parts:

$$f(x) = 4 - 3x$$

$$-3x + 4 = 4 + \sum_{n=1}^{\infty} \frac{6(-1)^n}{n} \sin(nx)$$

(d) $f(x) = e^x$

$$e^x = \frac{e^{\pi} - e^{-\pi}}{2\pi} + \sum_{n=1}^{\infty} \frac{(e^{\pi} - e^{-\pi})(-1)^n}{\pi(1 + n^2)} \cos(nx) - \sum_{n=1}^{\infty} \frac{n(e^{\pi} - e^{-\pi})(-1)^n}{\pi(1 + n^2)} \sin(nx)$$

(e) $f(x) = \tan(x)$

$\tan(x)$ has vertical asymptotes in $[-\pi, \pi]$, so:

No standard Fourier series exists on $[-\pi, \pi]$

2. Half-wave symmetry:

- (a) Yes
- (b) Yes
- (c) No
- (d) No

3. Half-wave symmetry:

- (a) No
- (b) Yes
- (c) Yes
- (d) No
- (e) Yes